Lin Ding Shan

AgriTech Computer Vision | Edge AI Deployment

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Kuala Lumpur, Malaysia · Available July 2027 · Relocating to New Zealand or Australia

PROFESSIONAL SUMMARY

Computer-vision engineer who takes detection models from orchard photos to working tools for growers. Founder of SnapDurian, a durian disease and pest scouting app that runs in a phone browser and started as a handheld Raspberry Pi 5 device; also built a disease-monitoring node for a commercial greenhouse (internship). Sole author of a paper on the device in *Smart Agricultural Technology* (Elsevier). Final-year Data Science student at UCSI University, graduating May 2027.

TECHNICAL SKILLS

Programming: Python, SQL, Bash; Git / GitHub
Computer vision: PyTorch, Ultralytics YOLO (v8, 11), RT-DETR, Faster R-CNN, MobileNet, EfficientNet, OpenCV
Deployment: ONNX Runtime, LiteRT / TFLite, INT8 and weight-only quantisation; Raspberry Pi 5, Jetson Orin Nano (CUDA), Linux, libcamera, V4L2
Web & backend: FastAPI, SQLite, Docker Compose, Caddy (HTTPS), PWA (service worker, offline queue), Telegram Bot API
Performance engineering: Latency, power and thermal telemetry; battery / UPS telemetry over I2C; hardware performance counters; duty cycling; multi-model scheduling
Data: Field data collection, annotation protocols, dataset merging and de-duplication

EXPERIENCE

Founder & Engineer | SnapDurian (snapdurian.com), pre-incorporation agri-tech venture, Malaysia *Jul 2025 – Present*

- Built SnapDurian as a phone web app (PWA): a grower photographs a leaf or trunk and sees the likely disease or pest among 11 categories marked on the photo, in English, Chinese or Malay; photos taken offline sync when signal returns.
- Run the backend: FastAPI and ONNX Runtime on a CPU cloud server (Docker, Caddy HTTPS), with invite codes and daily limits for grower trials, a tiled second pass for small lesions in distant shots, model swaps without restart, and one-tap grower feedback exported as pre-labels for retraining.
- Photographed a farm-labelled dataset at 12 orchard sites in Peninsular Malaysia and Sabah from Dec 2025; agreed the lesion-level annotation standard with a crop-protection practitioner, managed outsourced annotation and reviewed every image. Wrote the merge pipeline for the product set (~2,000 images, ~23,000 lesion boxes), keeping duplicate photos within one split.
- Started as an offline handheld Raspberry Pi 5 device and moved to the phone app after farm tests showed the kit was awkward to carry and overheated in a backpack. On the Pi, one unified YOLO11s model replaced two detectors, cutting response time **from 499 to 351 ms (−30%)** with no loss of accuracy.
- Screened INT8 on four backbones: 2.9–4.2× faster, but **it pushed the two smallest-target pest categories below a usable level on every backbone**; weight-only quantisation kept 96–99% of FP32 accuracy at 24–40% lower energy per image.
- On the Pi, traced monitoring gaps to the 82 °C software thermal cut-off, not throttling (2.6% latency cost), and removed them in lab tests with a 30 s duty-cycle sleep; running two detectors in turn instead of at once avoided a 58–150% slowdown.
- Benchmarked a Jetson Orin Nano Super (pest detection 186 → 35 ms, at 2.9× the cost and ~1.4× the power) and measured that idle draw dominates the Pi (6.3 W idle vs 8.4 W inferring).

AI Development Intern | Urban Farm Tech, Malaysia *Jan 2026 – May 2026*

Sole AI engineer as the IoT firm moved into AI.

- Delivered an offline basil disease-monitoring node for a commercial greenhouse: MobileNetV2 trained on ~1,700 on-site images and exported to ONNX on Raspberry Pi 5, with Telegram alerts confirmed over 3-of-5 frames in under 2.5 s.
- Traced a 46% Docker-vs-native latency gap to the stale ONNX Runtime in the image (1.19 vs 1.29), not Docker (a later controlled study put Docker's own cost at 1.3 ms); wrote an environment-fingerprint tool to keep deployments reproducible.

PUBLICATIONS & OPEN DATA

Lin, D. S. (2026). Thermal cut-off, not frequency throttling: duty-cycle scheduling and a dataset-integrity audit for on-farm edge disease detection in durian. *Smart Agricultural Technology* 15, 102526. doi:10.1016/j.atech.2026.102526. Sole author.

Three further sole-author manuscripts submitted: unified vs multi-model edge pipelines, release scheduling for two detectors on one board, and container overhead on edge nodes.

Datasets and measurement code are public on Zenodo (with DOIs) and GitHub; links on the portfolio.

EDUCATION

Bachelor of Computer Science in Data Science with Honours | UCSI University *Sep 2024 – May 2027 (expected)*

Languages: English (fluent), Mandarin (native) · **Work rights:** Malaysian citizen; visa sponsorship required (NZ, AU)